

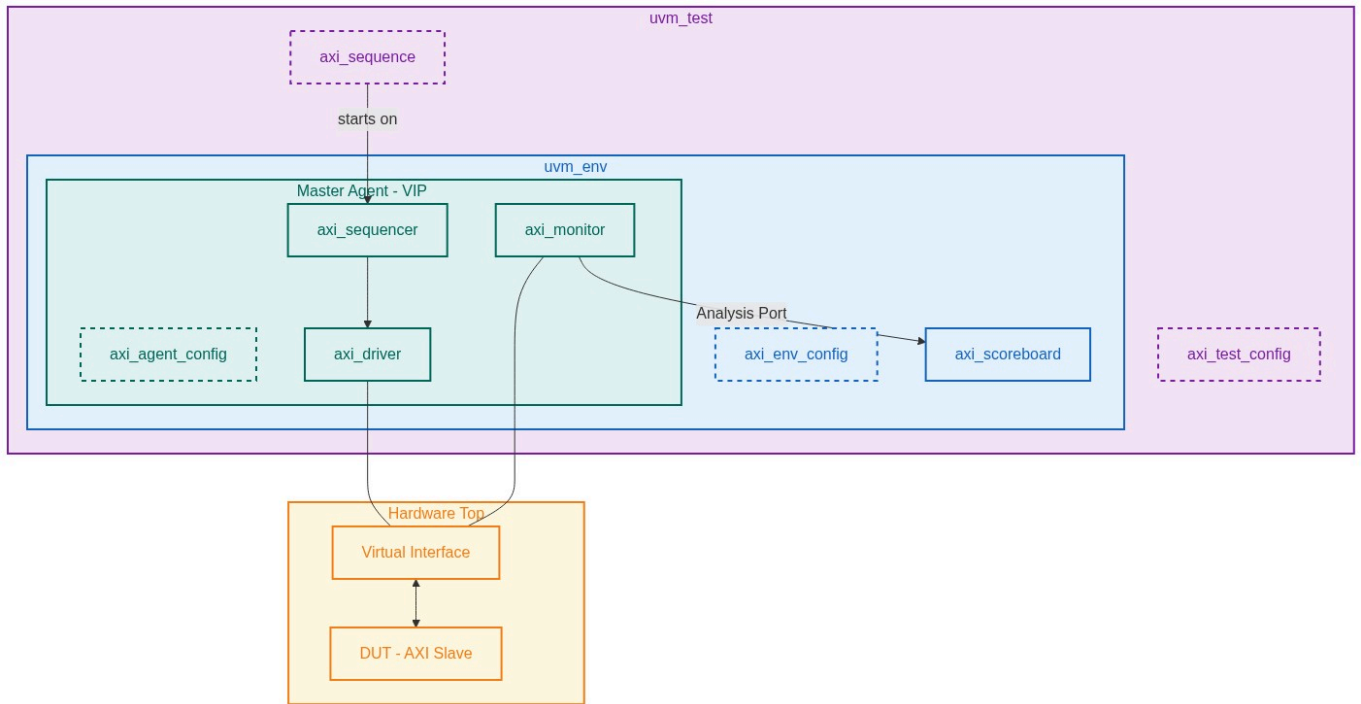
Verification Plan Report

Project: AXI-Stream Slave DUT Verification Plan ? VIP as Master (Transmitter)

Version: 5.0

Verification Role: Slave

1. UVM Environment Architecture



2. Protocol Configurations

Feature	Configuration Name	Impact
Handshake Protocol	Bidirectional Flow Control via TVALID/TREADY	The two-way handshake mechanism is the foundational conformance boundary of the AXI-Stream interface. The VIP (Master Agent) drives TVALID and payload, while the DUT Slave asserts TREADY to admit transfers. Deterministic latency compliance requires that the scoreboard detect any sampling anomaly ? specifically, data consumption occurring outside the coincident HIGH state of both signals.

Verification Plan Report

Packet Framing	TLAST Boundary Delineation	TLAST acts as the sole explicit packet-boundary indicator. Its assertion count must be preserved between Transmitter and Receiver without exception. The DUT's internal state machines are expected to flush on TLAST=1; the monitor must reconstruct complete datagram sequences and verify that inter-packet state isolation is maintained.
Byte Qualifiers	TKEEP / TSTRB Null and Position Byte Encoding	TKEEP qualifies bytes for transport (null byte elimination) while TSTRB distinguishes data bytes from positional placeholders. The reserved combination (TKEEP=0/TSTRB=1) is an illegal stimulus that must be monitored via protocol assertion. Correct Slave handling of sparse streams requires byte-level reconstruction fidelity in the scoreboard.
Parity Protection (AXI5)	Odd Parity ? Check_Type: Odd_Parity_Byte_All	The AXI5-Stream parity scheme provides single-bit error detection across all interface signals. The VIP Master must generate TDATACHK, TVALIDCHK, and TLASTCHK on every qualifying cycle per Table 5-1. The DUT Slave is expected to propagate detection events for deliberate fault-injection scenarios, verifying error-containment behavior.
Reset Sequencing	ARESETn ? Synchronous Deassertion Protocol	ARESETn is asserted asynchronously but must be deasserted synchronous to ACLK. The DUT Slave must not latch any transfer during ARESETn asserted state and must hold its state-machine at the designated IDLE state until the first valid post-reset clock edge. Mid-transfer reset injection tests the DUT's ability to suppress in-flight data without residual coherency corruption.

Verification Plan Report

Wakeup Signaling (AXI5)	TWAKEUP Pre-Transfer Activity Assertion	In power-aware topologies, TWAKEUP must be asserted at minimum one cycle before TVALID. The Slave is permitted to gate TREADY until TWAKEUP is detected, and doing so without deadlock is a key conformance requirement. Verification must exercise both the recommended (early) and permissible (simultaneous) wakeup timing windows.
Transfer Interleaving	Per-Transfer TID-Indexed Stream Interleaving	Interleaving is permitted on a per-transfer granularity across distinct TID/TDEST stream identifiers. The DUT Slave must maintain orthogonal stream contexts, ensuring that data integrity and ordering are preserved independently for each unique identifier pair. Scoreboard stream demultiplexing logic must validate this isolation property.
Continuous Packets Mode	Continuous_Packets Property Enforcement	When Continuous_Packets=True, the protocol mandates that TSTRB is absent, all mid-packet TKEEP bits be HIGH, and TID/TDEST remain stable while TLAST=0. Arbitration is constrained to TLAST boundaries. VIP sequences operating in this mode must enforce these constraints structurally, and the protocol monitor must flag any mid-packet TID mutation as a conformance violation.

3. Test Requirements

Index	Name	Description
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Verification Plan Report

REQ_SLV_01	TREADY Conformance and Handshake Latency	The DUT Slave drives TREADY to signal its readiness to accept an inbound transfer. A transfer is architecturally consummated only when both TVALID and TREADY are simultaneously asserted on a rising ACLK edge. The Slave is permitted to assert TREADY before TVALID is observed, enabling zero-overhead acceptance paths. However, it is also permitted to gate TREADY until TVALID is seen, introducing a deterministic, observable latency. Verification Intent: The Master Agent must exercise all three canonical handshake orderings (TVALID-first, TREADY-first, simultaneous) across constrained-random sequences. A scoreboard assertion must verify that no TDATA sampling event occurs outside a coincident dual-assertion window. Back-pressure injection via randomized TREADY stalls must exercise the Slave's ability to sustain flow control without internal FIFO overflow or state corruption.
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Verification Plan Report

REQ_SLV_02	Packet Boundary Integrity via TLAST Acknowledgment	The TLAST signal demarcates the terminal beat of each AXI-Stream packet. Its assertion count must be conserved end-to-end; it is not permissible for the Slave to suppress or duplicate TLAST events. Additionally, the Slave must correctly distinguish packet-start context (the first transfer following a prior TLAST assertion) from mid-packet context. The boundary between two consecutive packets has no explicit start signal; it is implicitly determined by the prior TLAST assertion and matching TID/TDEST identifiers. Verification Intent: Sequences must cover single-beat packets (TLAST=1 on beat 0), maximum-length variable-beat packets, and back-to-back packet streams with no idle gap between TLAST=1 and the subsequent TVALID=1. The monitor must independently reconstruct packets from observed beat sequences and the scoreboard must verify correctness of inter-packet state isolation within the DUT.
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Verification Plan Report

REQ_SLV_03	Byte	Qualifier	Compliance ?	The TKEEP and TSTRB signals encode per-byte semantics for each TDATA beat. TKEEP=1/TSTRB=1 denotes a data byte; TKEEP=1/TSTRB=0 denotes a positional placeholder byte; TKEEP=0/TSTRB=0 denotes a null byte eligible for removal. The combination TKEEP=0/TSTRB=1 is architecturally reserved and constitutes a protocol violation. The DUT Slave must correctly interpret the byte stream without altering the relative ordering of data and positional bytes. Verification Intent: The Master Agent must drive all four TKEEP/TSTRB permutations under constrained randomization, with the reserved combination (TKEEP=0/TSTRB=1) used exclusively for negative-stimulus injection. A real-time protocol assertion on the monitor interface must flag this illegal encoding. Sparse-stream sequences with statistically distributed null bytes must verify that the Slave's internal data reconstruction is tolerant of non-contiguous active bytes across wide-bus configurations.
	TKEEP/TSTRB		Semantic	
	Interpretation			

Verification Plan Report

REQ_SLV_04	Reset Conformance ? ARESETn Synchronous Deassertion and State Recovery	ARESETn governs the Slave's initialization state transitions. The signal may be asserted asynchronously at any point in the transfer timeline, but its deassertion must be synchronous to the rising edge of ACLK. During the assertion window, the Slave must unconditionally hold all internal state machines at the defined IDLE condition. Any in-flight data present at the interface boundary must be suppressed without residual coherency effects or FIFO saturation. Verification Intent: Reset stimulus must be injected across three distinct interface states: (1) idle, (2) mid-packet with TVALID=1 and TREADY gated LOW, and (3) at the moment of handshake coincidence (TVALID=TREADY=1). Post-reset recovery sequences must confirm that the DUT correctly resumes transfer acceptance on the first eligible ACLK edge following ARESETn deassertion, with no residue of the pre-reset transaction state.
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Verification Plan Report

REQ_SLV_05	AXI5 Parity Fault Detection on Inbound TDATACHK	The AXI5-Stream interface parity scheme employs odd parity checking across all interface signals. TDATACHK provides TDATA_WIDTH/8 check bits, one per byte lane. Parity must be driven correctly in every cycle in which its associated Check Enable condition is asserted. Critically, parity must be generated for ALL byte lanes regardless of whether the associated TKEEP bit is asserted, precluding any null-byte optimization in parity generation. Verification Intent: The Master Agent must generate correctly computed TDATACHK, TVALIDCHK, TLASTCHK, and TREADYCHK on every eligible cycle. Fault injection scenarios must introduce deliberate single-bit inversions on specific check-bit positions and verify that the DUT Slave activates its error-reporting path. The scoreboard must independently compute expected parity and assert equivalence on every monitored beat to achieve 100% check-bit coverage.
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Verification Plan Report

REQ_SLV_06	TWAKEUP Pre-Transfer Activity Signaling (AXI5)	The TWAKEUP signal serves as a power-domain activity indicator that precedes active data transfers. The specification recommends its assertion at least one ACLK cycle before TVALID. If asserted concurrently with TVALID, TWAKEUP must sustain its assertion until TREADY is also observed HIGH. The Slave is architecturally permitted to defer TREADY until TWAKEUP is detected, which introduces a potential deadlock condition if TWAKEUP is never driven. Verification Intent: Timing stimulus must cover: (a) TWAKEUP leading TVALID by one or more cycles; (b) simultaneous assertion of TWAKEUP and TVALID; and (c) degenerate cases where TWAKEUP is toggled without a subsequent transfer, verifying that the Slave does not incorrectly advance its internal ready state. Deadlock watchdog logic in the test must detect and flag scenarios where the Slave holds TREADY indefinitely.
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Verification Plan Report

REQ_SLV_07	TID/TDEST Stream Identifier Ordering Orthogonality	The AXI-Stream protocol enforces strict transfer ordering: reordering of beats within any given TID/TDEST stream is prohibited. Transfer interleaving at per-beat granularity is permitted across distinct stream identifiers, enabling multiplexed transport on a shared physical interface. The DUT Slave must maintain independent data-path contexts for each active TID/TDEST pair and must correctly demultiplex interleaved beats without cross-stream contamination. Verification Intent: Multi-stream sequences must interleave transfers across at least two distinct TID values. The scoreboard must independently track per-stream byte sequences using TID/TDEST as demultiplexing keys and verify intra-stream ordering preservation. Protocol assertions must flag any observable reordering event. Stress scenarios must sustain maximum permissible interleaving depth to characterize Slave context-switching capacity.
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Verification Plan Report

REQ_SLV_08	Null Packet Reception ? Zero-Data TLAST Transfer Handling	A null packet is defined as a transfer where TLAST=1 is asserted but all TKEEP bits are deasserted (TKEEP=all-zeros), resulting in a packet containing no data or positional bytes. This construct is used to indicate packet-end conditions, flush intermediate buffers, or satisfy endpoint synchronization requirements without payload. The Slave must acknowledge the null packet via handshake completion without triggering unintended data path operations. Verification Intent: Master Agent must generate null packets at packet boundaries, mid-stream (following a non-null packet), and in back-to-back sequences. The scoreboard must verify that the Slave's internal byte counter is not incremented by null packet events, and that the Slave's packet-boundary state machine correctly transitions to the next-packet context without treating the null TLAST as a data-bearing frame.
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4. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_SLV_001	TREADY-First Handshake: The VIP Master drives TVALID LOW for N cycles (1 to 8, constrained random) while the Slave has pre-asserted TREADY. Upon TVALID assertion, the transfer must complete in the same cycle. Verify DUT does not latch undefined data during the pre-TVALID window.	REQ_SLV_01
TC_SLV_002	TVALID-First Handshake with Back-Pressure: Master asserts TVALID and holds data stable. DUT Slave deasserts TREADY for a randomized stall period (0 to 32 cycles). Scoreboard verifies that TDATA, TKEEP, TLAST, TID, and TDEST remain bit-stable across the entire stall window per Section 2.2.	REQ_SLV_01

Verification Plan Report

TC_SLV_003	Sustained Back-Pressure Stress: Master drives a 256-beat packet with TREADY toggling at a 50% duty cycle. Scoreboard reconstructs the complete packet from monitored beats and validates payload integrity against the expected reference model. Measures effective throughput efficiency.	REQ_SLV_01, REQ_SLV_02
TC_SLV_004	Single-Beat Packet: TVALID=1, TLAST=1 asserted simultaneously on the first and only beat. Slave must consummate the handshake and transition to inter-packet idle state within one clock cycle of TREADY assertion. Scoreboard checks that TLAST count increments by exactly one.	REQ_SLV_02
TC_SLV_005	Back-to-Back Packet Stream: TVALID remains HIGH across the TLAST boundary ? the last beat of packet N (TLAST=1) is immediately followed by the first beat of packet N+1 (TLAST=0) with no idle gap. Verify DUT correctly partitions the two packets and does not merge beats across the boundary.	REQ_SLV_02
TC_SLV_006	Sparse Byte-Qualifier Stream: Master drives TKEEP/TSTRB with constrained-random per-byte patterns across a 4-byte data bus, covering all valid encoding combinations (data, position, null bytes). Scoreboard verifies byte ordering and data reconstruction integrity. Monitor asserts that TKEEP=0/TSTRB=1 (reserved) never appears.	REQ_SLV_03
TC_SLV_007	Reserved Qualifier Injection (Negative Stimulus): Master injects a single beat with TKEEP=0, TSTRB=1 on one byte lane. The VIP protocol monitor must raise a fatal assertion. DUT behavior post-injection is observed and logged; no system deadlock should occur.	REQ_SLV_03
TC_SLV_008	Mid-Transfer Reset Injection: While a multi-beat packet is in-progress (TVALID=1, TREADY=0), ARESETn is driven LOW for 4 ACLK cycles. Verify DUT deasserts TREADY within one cycle of reset assertion and internal FIFOs are in a flushed state. Post-reset, confirm that the first new transfer is accepted correctly.	REQ_SLV_04
TC_SLV_009	Reset During Handshake: ARESETn deasserted precisely at the ACLK edge where TVALID=TREADY=1 (handshake boundary). Verify that the transfer is not partially sampled and that the DUT returns cleanly to IDLE without asserting a spurious completion event.	REQ_SLV_04
TC_SLV_010	AXI5 Parity Validation ? Nominal Traffic: VIP Master drives correct odd parity on TDATACHK, TVALIDCHK, and TLASTCHK for all beats across a 1000-transfer regression. Scoreboard independently computes expected parity and asserts bit-exact equivalence on every beat.	REQ_SLV_05

Verification Plan Report

TC_SLV_011	AXI5 Parity ? Null Byte Lane Coverage: With TKEEP=0 on two of four byte lanes, VIP must still drive correct TDATACHK bits for those inactive lanes. Scoreboard verifies that parity is computed over all TDATA bits regardless of TKEEP state.	REQ_SLV_05
TC_SLV_012	AXI5 Parity Fault Injection: Master deliberately inverts one bit of TDATACHK on beat index 47. Verify that the DUT Slave activates its error-reporting mechanism (error flag output, error interrupt, or protocol assertion) within the cycle of detection.	REQ_SLV_05
TC_SLV_013	TWAKEUP Recommended Timing: TWAKEUP asserted exactly 2 cycles before TVALID. Verify Slave accepts TREADY assertion within the specified wakeup latency budget. No deadlock must occur.	REQ_SLV_06
TC_SLV_014	TWAKEUP Simultaneous with TVALID: TWAKEUP and TVALID asserted in the same cycle. Verify TWAKEUP remains HIGH until TREADY is asserted, and that the Slave does not exhibit TREADY starvation.	REQ_SLV_06
TC_SLV_015	Dual-Stream Interleaving: Two concurrent streams (TID=0, TID=1) are interleaved at transfer granularity. The VIP drives alternating beats from each stream across a 64-beat sequence. Scoreboard demultiplexes by TID and independently validates ordering for each stream without cross-stream data contamination.	REQ_SLV_07
TC_SLV_016	Null Packet Reception: VIP drives TVALID=1, TLAST=1, all TKEEP=0 on a single beat. Slave must assert TREADY and complete the handshake. Scoreboard verifies no data bytes are credited to the null packet and the DUT correctly advances to inter-packet idle state.	REQ_SLV_08
TC_SLV_017	Back-to-Back Null Packets: Two consecutive null packets (TLAST=1, TKEEP=all-0) with no idle gap between them. Verifies the Slave's sequential packet-boundary state machine does not lock up on zero-data transitions.	REQ_SLV_08, REQ_SLV_02

5. Functional Coverage Groups

Group Name	Description
cg_handshake_ordering	Three-bin cross of handshake ordering scenarios: (a) TVALID before TREADY, (b) TREADY before TVALID, (c) simultaneous assertion. Cross with TREADY stall depth bins: 0, 1-4, 5-16, 17-32 cycles.
cg_packet_size_distribution	Packet-length bins: 1 beat (single), 2-4 beats (short burst), 5-32 beats (medium), 33-256 beats (long). Separate bins for back-to-back and isolated packet categories.

Verification Plan Report

cg_byte_qualifier_patterns	Per-byte lane TKEEP/TSTRB encoding coverage. Bins: all-data, all-null, mixed (data+null), position-byte present, sparse (randomized). Reserved combination coverage must be zero (functional) and non-zero only in negative-stimulus scenarios.
cg_reset_injection_state	Three-state reset injection coverage: (a) idle, (b) mid-packet TVALID=1/TREADY=0, (c) handshake coincidence TVALID=TREADY=1. Cross with reset duration bins: 1, 2-4, 5-16 cycles.
cg_parity_check_lanes	Per-byte-lane parity bit coverage across all TDATACHK positions for a 4-byte bus. Separate bins for active (TKEEP=1) and inactive (TKEEP=0) lanes. Fault injection coverage: bit-0 through bit-3 of TDATACHK individually inverted.
cg_twakeup_timing	TWAKEUP lead-time bins relative to TVALID: -1 cycle (post-TVALID, degenerate), 0 cycles (simultaneous), 1 cycle (recommended), 2-4 cycles (early). Cross with transfer duration.
cg_stream_interleaving_depth	Number of active TID streams: 1, 2, 4. Consecutive transfer count per TID before stream switch: 1 (maximum interleaving), 2-8, 9+. Cross verifies Slave context-switching saturation boundary.
cg_null_packet_context	Null packet occurrence context bins: isolated (no surrounding packets), following a data packet, preceding a data packet, back-to-back null packets. Verifies TLAST event handling orthogonality.

6. Interface Signals

Signal Name	Direction	Width	Description
ACLK	Input (to DUT Slave)	1	Global synchronization clock. All input signals are sampled on the rising edge. All output signal changes from the Slave must occur after the rising edge. (Section 2.8.1, p.28)
ARESETn	Input (to DUT Slave)	1	Active-LOW global reset. May be asserted asynchronously at any time. Deassertion is synchronous to the rising edge of ACLK. Mandates TREADY=0 from the Slave during assertion. (Section 2.8.2, p.28)
TVALID	Input (VIP Master Output)	1	Indicates that the VIP (Transmitter) is driving a valid transfer payload. Once asserted, must remain HIGH until the handshake is completed. (Section 2.2, p.18)
TREADY	Output (DUT Slave Output)	1	Indicates the DUT Slave can accept an inbound transfer. May be pre-asserted or deferred until TVALID is observed. Default value is HIGH if signal is absent from the interface. (Section 3.1.1, p.34)
TDATA	Input (VIP Master Output)	TDATA_WIDTH (integer x8 bits)	Primary payload signal. Encodes the ordered byte sequence of the data stream. Must remain stable for the full duration of any TVALID assertion window where TREADY has not yet been asserted. (Section 2.4, p.21)

Verification Plan Report

TSTRB	Input (VIP Master Output)	TDATA_WIDTH /8	Per-byte qualifier distinguishing data bytes (TSTRB=1) from positional placeholder bytes (TSTRB=0). Only meaningful when associated TKEEP bit is asserted. (Section 2.5.2, p.24)
TKEEP	Input (VIP Master Output)	TDATA_WIDTH /8	Per-byte null-byte qualification signal. When deasserted for a given byte lane, that lane carries a null byte eligible for removal by the Slave or downstream interconnect. Default: all-HIGH if signal is absent. (Section 2.5.1, p.24)
TLAST	Input (VIP Master Output)	1	Packet boundary indicator. Asserted on the terminal beat of each packet. TLAST assertion count must be preserved without modification between Transmitter and Receiver. (Section 2.6, p.26)
TID	Input (VIP Master Output)	TID_WIDTH (recommended <= 8 bits)	Stream identifier enabling per-transfer multiplexing of independent data streams on a shared physical interface. Transfers sharing TID/TDEST belong to the same stream and must maintain in-order arrival. (Section 2.7, p.27)
TDEST	Input (VIP Master Output)	TDEST_WIDTH (recommended <= 8 bits)	Routing identifier providing coarse destination addressing for stream-level routing within interconnect fabrics. Merging of transfers with differing TID or TDEST is prohibited. (Section 2.7, p.27)
TUSER	Input (VIP Master Output)	TUSER_WIDTH (recommended: integer multiple of TDATA_WIDTH /8)	User-defined sideband signal aligned on a per-byte basis to TDATA. Byte x occupies TUSER[((x*m)+(m-1)):(x*m)]. TUSER values for null bytes (TKEEP=0) are not guaranteed to be transmitted. (Section 2.9, p.29)
TWAKEUP	Input (VIP Master Output)	1 (AXI5-Stream only)	Activity indicator for power-domain management. Must be glitch-free and is recommended to be asserted at least one cycle before TVALID. Slave may defer TREADY until TWAKEUP is observed. Present only when Wakeup_Signal property is True. (Section 2.3, p.20)
TVALIDCHK	Input (VIP Master Output)	1 (AXI5-Stream only)	Odd parity check bit for TVALID. Check Enable: ARESETn=1. Driven correctly every cycle while the enable condition holds. (Table 5-1, p.48)
TDATACHK	Input (VIP Master Output)	TDATA_WIDTH /8 (AXI5-Stream only)	Odd parity check field for TDATA. One bit per 8-bit lane. Check Enable: TVALID=1. Must be driven correctly for ALL byte lanes including lanes where TKEEP=0. (Table 5-1, p.48; Section 5.3, p.46)

Verification Plan Report

TLASTCHK	Input (VIP Master Output)	1 (AXI5-Stream only)	Odd parity check bit for TLAST. Check Enable: TVALID=1. The parity bit for a 1-bit signal is its logical inversion, as per the critical control signal rule of Section 5.3. (Table 5-1, p.48)
TREADYCHK	Output (DUT Slave Output, AXI5-Stream only)	1	Odd parity check bit for TREADY. Generated by the DUT Slave. Check Enable: ARESETn=1. Scoreboard must validate this output against the independently computed expected parity. (Table 5-1, p.48)

7. Verification Environment Structure

Root: axi_stream_slave_vip_tb/

- |-- master_agent : VIP Transmitter Agent: Sequencer, Driver (TVALID/TDATA stimulus), Monitor
- |-- env : UVM Environment: Protocol Scoreboard, Coverage Subscriber, Config Object
- |-- sequences : Stimulus library: handshake, sparse-stream, back-to-back, null-packet, parity, reset, interleaving
- |-- tests : UVM test classes mapping scenarios to sequence combinations
- |-- assertions : SVA property file for real-time protocol conformance monitoring
- |-- top : Simulation top-level: DUT instance, virtual interface binding, clock/reset generation
- |-- verif_plan : Verification plan artifacts: YAML matrix, PDF report, sub-prompt references